



Series · Worksheet 1

Core Pure 1 Chapter 3: Series. Answer all questions. Show your working clearly. Total: 65 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- You may quote the standard results for $\sum r$, $\sum r^2$ and $\sum r^3$ unless told otherwise. Give exact answers, with algebraic answers fully factorised.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Standard summation formulae	18	
B Sums over a range and factorised forms	21	
C Method of differences	21	
Roar-level challenge	5	
Total	65	

Section A Standard summation formulae

18 marks

1. Find the value of

[2]

(a) $\sum_{r=1}^{20} r,$

(b) $\sum_{r=1}^{10} r^2.$

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2. Find the value of $\sum_{r=1}^{12} r^3.$

[2]

.....

.....



3. Find the value of $\sum_{r=1}^{15} (3r^2 - 2r)$.

[3]

4. Show that

[4]

$$\sum_{r=1}^n (6r^2 + 2r) = 2n(n+1)^2.$$



5. (a) Show that, for all positive integers n ,

[4]

$$\sum_{r=1}^n r(r+2) = \frac{1}{6}n(n+1)(2n+7).$$

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(b) Hence find the positive integer n for which

[3]

$$\sum_{r=1}^n r(r+2) = 4n^2.$$

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(Total for Question 5 is 7 marks)



Section B Sums over a range and factorised forms

21 marks

6. Find the value of $\sum_{r=11}^{20} r$. [2]

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7. Find the value of $\sum_{r=10}^{20} r^3$. [3]

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8. Show that [3]

$$\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(7n+1).$$

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9. Show that [4]

$$\sum_{r=1}^n (r+1)(r+3) = \frac{1}{6}n(2n^2 + 15n + 31).$$

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10. (a) Show that, for all positive integers n ,

[4]

$$\sum_{r=1}^n r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3).$$

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(b) Hence show that

[3]

$$\sum_{r=n+1}^{2n} r(r+1)(r+2) = \frac{3}{4}n(n+1)(an^2 + bn + c),$$

where a , b and c are integers to be found.

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(c) Hence find the value of $\sum_{r=11}^{20} r(r+1)(r+2)$.

[2]

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(Total for Question 10 is 9 marks)



Section C Method of differences

21 marks

11. Express $\frac{2}{r(r+2)}$ in partial fractions. [2]

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12. Using $\frac{1}{r(r+1)} \equiv \frac{1}{r} - \frac{1}{r+1}$, show that [3]

$$\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}.$$

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13. Given that $f(r) = r^2(r+1)^2$, show that $f(r) - f(r-1) = 4r^3$. [3]
Hence use the method of differences to prove that $\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$.

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14. Use the method of differences to show that

[4]

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}.$$

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15. (a) Express $\frac{1}{r(r+2)}$ in partial fractions.

[2]

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(b) Hence show that

[4]

$$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{n(3n+5)}{4(n+1)(n+2)}.$$

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(c) Find the exact value of $\sum_{r=11}^{20} \frac{1}{r(r+2)}$.

[2]

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(d) Write down the limit of $\sum_{r=1}^n \frac{1}{r(r+2)}$ as $n \rightarrow \infty$.

[1]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Show that $\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \equiv \frac{2}{r(r+1)(r+2)}$. [5]

Hence find $\sum_{r=1}^n \frac{1}{r(r+1)(r+2)}$ as a single fraction in its simplest form, and deduce the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r+2)}.$$

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END OF WORKSHEET

Total: 65 marks